

JEE MAIN REVISION MASTER: P-BLOCK ELEMENTS SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 254 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	P Block Elements - Core Application	254 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 P Block Elements - Core Application (Very High Yield — 254 PYQs)

Core Principle: Key problem pattern in P Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Find the correct statements related to group 15 hydrides. A. Reducing nature increases from NH_3 to BiH_3 B. Tendency to donate lone..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Correct statements from the following are: A. Nitrogen in oxidation states from +1 to +4 disproportionates in acid medium. B. Nitrogen has t..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I: The second ionization enthalpy of B, Al and Ga is in the order of $B > Al > Ga$. Statement II: T..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#2)

Treatment of a gas 'X' with a freshly prepared ferrous sulphate solution gives a compound 'Y' as a brown ring. The compounds X and Y are.

- (A) NO and $[\text{Fe}(\text{NO})]\text{SO}_4$ **[Correct]**
- (B) NO_2 and $[\text{Fe}(\text{NO}_2)]\text{SO}_4$
- (C) N_2O and $[\text{Fe}(\text{N}_2\text{O})]\text{SO}_4$
- (D) N_2O_4 and $[\text{Fe}(\text{N}_2\text{O}_4)]\text{SO}_4$

Correct Answer: A

Solution Analysis: The reaction described is the well-known brown ring test, which is a confirmatory test for nitrate ions (NO_3^-). Let's understand the chemistry behind it. Understanding the Reactants: The test uses a freshly prepared solution of ferrous sulphate (FeSO_4). It needs to be fresh because the ferrous ion (Fe^{2+}) can be easily oxidized to ferric ion (Fe^{3+}) by the oxygen in the air. The question mentions a gas 'X' is treated with this solution. In a standard brown ring test, this gas is produced in situ (within the reaction mixture). Let's see how. Formation of Gas 'X':